Intelligent Weather Classification Using Deep Learning

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Abstract— Image classification of weather is key to applications like transportation, agriculture and environmental monitoring. The existing traditional methods of sensor and manual observation are not always flexible and adaptive in real-time and require the extensive use of human resources. This paper suggests image-based weather classification based on deep learning of which EfficientNet-B3 will be used as the main evaluation tool and compared with EfficientNet-B0, DenseNet121 and EfficientNetV2-S. The multi-class weather condition is trained on a set of five weather conditions in the model, i.e., cloudy, foggy, rainy, snowy, and sunny. As the experimental findings suggest, EfficientNet-B3 obtains the accuracy of classifying images 93.54 percent and its performance remains more stable to image distortions, such as blurred images or color changes. The results demonstrate the suitability of EfficientNet based designs in both precise and robust weather categorization in practice. In numerous real-life tasks, weather classification is a significant activity, including transportation, agriculture, as well as environmental monitoring.

Keywords—*weather image classification, convolutional neural networks, transfer learning, computer vision, insert*

1. Introduction

Weather forecasting enables accurate identification of weather conditions to support planning and decision-making. With the advancement of computer vision, image-based weather classification has emerged as an effective alternative to traditional sensor-based systems. Efficient deep learning architectures such as EfficientNet improve image classification performance by balancing network depth, width, and resolution using a compound scaling method [1]. Furthermore, several studies have demonstrated the effectiveness of convolutional neural networks for weather image classification tasks [2].

It's really tough to figure out what the weather is like just by looking at pictures. This is because rainy, foggy, and cloudy conditions can look pretty similar, and things like lighting, image quality, and the environment can make it even harder to tell. Older methods that use special features and machine learning algorithms often have a hard time dealing with all these differences. Even though newer deep learning models are better at getting the right answer, most research only looks at how accurate they are and doesn't check how well they work with real-world pictures that can be blurry or have other issues.

To address these limitations, this study proposes a deep learning-based weather classification system using EfficientNet-B3 as the primary model. In addition, other models such as EfficientNet-B0, DenseNet121, and EfficientNetV2-S are used for comparative analysis. The proposed approach also evaluates model robustness by testing under different image conditions such as blur and color transformations.

The findings show that models using EfficientNet do a great job of classifying weather, and the new system is more stable when dealing with different image conditions. This is really useful for real-world situations where the quality of images and the environment can change a lot. It means the approach can handle varying conditions and still provide good results, making it more practical for everyday use.

2. WEATHER IMAGE AND FEATURE EXTRATION ANALYSIS

When looking at pictures of the weather, it's not just about seeing the difference between a sunny day and a rainy one. We need to find the tiny details that set each kind of weather apart. This means looking at the colors, textures, and how bright or dark the picture is. We also have to consider the patterns we see in the images. But real-life weather pictures can be tricky - they might be blurry, noisy, or the lighting could be off. So, we have to be clever about how we analyze these pictures and find the strong features that will help us tell the difference between them. To do this, we use a special tool called EfficientNet-B3, which can automatically find these features for us, making it easier to classify the weather in the pictures and see how well our method works.

2.1 Analysis of Weather Image

When we look at pictures of the weather, we can see that they are all different. Some are cloudy, some are foggy, some are rainy, some are snowy, and some are sunny. These pictures don't just look different, they also have different properties. For example, the colors in them are spread out in different ways, the textures are different, and the lighting is different too. If we compare some of these pictures, like the foggy and cloudy ones, we can see that they look pretty similar. But if we compare others, like the sunny and snowy ones, we can see that they are really different, especially when it comes to how bright and clear they are.

When we look at real-world weather images, they can be a bit fuzzy or have color issues. This can make it hard to see important things. So, the model we use to classify these images needs to be good at finding features that don't change even when the image is fuzzy or the colors are off. It's like trying to recognize a picture even if it's not perfectly clear. The model has to be able to see past the problems and find the important stuff. This way, it can still give us good information about the weather, even if the image isn't perfect. [1], [2]

2.2 Motivation

Most existing weather classification approaches primarily focus on classification performance and do not analyze how model behavior changes under varying image conditions. In real-world scenarios, images may contain noise, blur, or changes in color representation, which can affect model predictions.

This project is driven by the need to assess not just how well a model can classify things, but also how robust it is. We chose EfficientNet-B3 because it's really good at learning detailed features from images. We also wanted to see how stable the model is, so we tested it with different versions of images, like blurry ones or ones in different color formats. This helps us understand how well the model can handle changes and still make good predictions. By doing this, we can get a better sense of how reliable the model is and whether it can work well in different situations.

2.3 Feature extraction

Feature extraction in this work is performed using the EfficientNet-B3 architecture, which automatically learns important visual features from input weather images. The model extracts hierarchical features using convolutional layers, enabling it to capture patterns relevant to different weather conditions.

2.3.1 Spatial Features**:** When we look at pictures of the weather, there are things like edges and shapes that help us understand what's going on. These are called spatial features. In a special kind of computer model called EfficientNet-B3, there are layers that use filters to find these features in the pictures. This is really helpful for seeing patterns like where the clouds are and where it's snowy. It's like the model is learning to recognize different kinds of weather, and it can tell them apart because of these patterns. The filters are applied all over the picture, which helps the model understand the whole scene and make sense of it.

$$F(i,j)=m\sum n\sum I(i+m,j+n)\cdot K(m,n) \qquad (1)$$

2.3.2 Texture Features: Texture features describe surface patterns present in images and are important for distinguishing visually similar weather conditions. For example, foggy images appear smooth, whereas rainy images may exhibit streak-like patterns. EfficientNet-B3 learns these texture patterns through deeper convolutional layers, improving class separation.

2.3.3 Color Features: Colors can tell us a lot about the weather. For example, when it's sunny, things tend to look brighter and more vibrant. On the other hand, when it's cloudy or foggy, everything can look dull and gray. A computer model was trained to recognize these patterns by looking at lots of pictures. The model learned to pick up on clues like bright colors when it's sunny and dull colors when it's not. But the question is, does the model really need all the color information to make good predictions? To find out, the model was tested using different ways of looking at colors, like HSV, LAB, and even just black and white. This helps us understand how much the model relies on color to figure out the weather.

2.3.4 Blur-based features: To see how well a model works, we use something called blur-based features. We add a Gaussian blur to the pictures at different levels - a little bit, a medium amount, and a lot - to make them look like they do in the real world when they're not perfectly clear. This helps us figure out if the model can still tell what the weather is like even when the picture isn't super sharp.

2.3.4 Color Space Transformation Features: Different color space transformations such as HSV, LAB, grayscale, BGR, and RGBA are applied during testing to analyze model sensitivity to color variations. This helps determine whether the model relies more on color information or structural features for classification.

3. RELATED WORK

EfficientNet Mingxing Tan and Quoc V. Le [1] provided a family of convolutional neural networks, EfficientNet, with a compound scaling to balance network depth, width, and resolution. The strategy is very effective in both the efficiency and accuracy of the models in contrast to the regular CNN designs. Nevertheless, the research is primarily based on generic image classification tasks and does not render its usefulness and strength in the image classification of weather in diverse real-world circumstances.

A study by Mingxing Tan and Quoc V. Le [2] introduced EfficientNetV2, which enhances the speed and efficiency of training and the efficiency of parameters with the help of training-aware neural architecture search and the progressive learning. The model is better performing at lower cost of computation. Nevertheless, the paper focuses on training effectiveness and performance is not compared when it comes to weather classification tasks and how images are manipulated based on model behaviour.

In [3], Mohammad Farid Naufal and Selvia Ferdiana Kusuma introduced an image classification model of weather in CNN with transfer learning. Several architectures, including MobileNetV2, VGG16, DenseNet and Xception were tested, and Xception delivered the highest performance. Nevertheless, it also concerns accuracy comparison, and the study does not pay much attention to such issues as robustness and work under different image conditions.

Harit Tarwani et al. [4] applied the transfer learning with pre-trained CNN models ResNet50V2, EfficientNetB5 and EfficientNetV2-S to classify weather. The models have been optimized to enhance the performance in various weather types. Despite the promising results of EfficientNetV2-S, the research lacks the examination of the model stability in case of any changes in images, which can include blur or color change.

Haixia Xiao et al. [5] introduced a deep CNN network that was called MeteCNN and generated weather phenomena category classification and presented a dataset of several weather categories. The model had shown a good performance in identifying different weather conditions. The study however concentrates on the classification performance and does not investigate the comparative analysis with the modern architectures and robustness analysis.

Vina Wulandari et al. [6] have compared CNN structures such as InceptionV3, DenseNet169 and NASNetMobile in the classification of weather images. The experiment compared the performance of the models based on various metrics and named the InceptionV3 as an optimal model. The work is however restricted to model comparison without exploring feature robustness in various image conditions.

Divya Pulipaka et al. [7] introduced a model of weather categorization with both standard machine learning and the deep learning models. The experiment did a combination of feature extraction with feature classifiers, including SVM, Random Forest, and CNN models. Nevertheless, the method uses handcrafted characteristics partially, and it does not exploit deep learning-based feature extraction and robustness analysis fully.

A CNN-based weather classification model developed by Ashish Sharma and Zaid Saad Ismail [8] is implemented on the Keras and TensorFlow. The model had high classification performance and proved that deep learning can be applied to weather prediction tasks. Nevertheless, the experiment is based on simple CNN design and does not compare with the in-depth models or study under different image parameters.

G. Lakshmi [9] suggested a deep CNN with batch normalization to distinguish weather. The model was very accurate and was able to classify a wide variety of weather conditions. However, the research has only a few comparative studies and has not assessed its robustness against various variations of images.

A weather classification scheme was proposed by Jose Carlos Villarreal Guerra, Zeba Khanam, Shoaib Ehsan, Rustam Stolkin, and Klaus McDonald-Maier [10] with a recently constructed multi-class dataset (RFS dataset) and data augmentation with the use of super pixel masking. The paper compared the various types of convolutional neural networks models in terms of extracting features and it was revealed that the deep learning networks could be successfully used to classify features like rain, fog, and snow. Nevertheless, the paper indicates average results and a lack of in-depth investigation of robustness in the face of real-life variations like lighting conditions and noise.

4. PROPOSED METHOD

This research presents a weather image classification system based on deep learning and EfficientNet-B3 as first. Besides, three trained models are employed as EfficientNet-B0, DenseNet121, and EfficientNetV2-S and utilized in the comparative analysis. The aim is to assess the classification performance, as well as model stability inequality, at different image conditions.

4.1 Flow of Proposed Model.

The objective of the suggested approach is to predict weather conditions based on the pictures with the help of the EfficientNet-B3 deep-learning framework. The workflow of the proposed system is shown in Fig. 1. The first step is the choice and pre-processing of weather images, in which the images are downsized to a constant input size and scaled to help ensure uniformity.

The dataset is preprocessed before being split into training and validation sets to facilitate the effective learning and appraisal. EfficientNet-B3 The base model is initialized with ImageNet pre-trained weights and employs them in extraction of features. Transfer learning is then used during training phase and fine tuning of the chosen layers is done in order to enhance representation of features that are unique to weather.

After the training of the model, it is tested on the prediction of weather classes. Moreover, robustness analysis is conducted by applying the variations of the images like Gaussian blur at various levels and color space conversions like HSV, LAB, and grayscale. This assists in testing stability of the model in various visual conditions.

Lastly, the features that have been extracted are inputted to the classification layer where they are categorized according to which of the five weather types each image belongs to. Critical metrics are used to measure the model performance that is analyzed to determine the classification capability and strength of the model.

4.2 Pre-trained model

This is known as pre-trained models, pre-trained models are models that have been trained on large data-set like ImageNet and can be applied to new tasks. This method saves on training time, enhances performance, and less labelled data is required. In this paper, the fine-tuning of pre-trained models is used to classify weather, which allows effective feature extraction and narrows the generalization gap.

4.2.1 EfficientNet-B0

EfficientNet-B0 is the base model of the EfficientNet family which mathematical basis is the compound scaling to normalize network depth, width and resolution [1]. It has several MBConv (Mobile Inverted Bottleneck Convolution) blocks with more efficient feature extraction with fewer parameters.

It takes the form of an architecture that starts with a convolutional layer then several MBConv blocks with varying expansion factors and kernel sizes. Each block has depthwise convolution, batch normalization and activation functions. It has a global average pooling layer that comes before the last fully connected classification layer.

EfficientNet-B0 is easily portable and computationally efficient and can be compared to highly sophisticated models. It has however limitations in the depth in which it can capture such intricate weather patterns.

4.2.2 DenseNet121

DenseNet121 is one such convolutional neural network, which is fed-forward, that is, it links each layer to every other layer. It is made of compression blocks and mixing layers, which contribute to efficient reuse of features and gradient flow [7].

The numerous convolutional layers in each dense block have feature maps joined and not added up. There exist transition layers that reduce dimensionality by use of convolution and pooling. It is terminated by the architecture with global average pooling then a fully connected classification network.

DenseNet121 is very effective in depicting detailed features, but it can use other computational resources than EfficientNet models

4.2.3 EfficientNetV2-S

EfficientNetV2-S is the enhanced adaptation of EfficientNet that aims at acquiring faster training and parameter-efficientnet [2]. It presents Fused-MBConv block, which is a concatenation of convolution operations to shorten the training time.

It is based on a mix of MBConv and Fused-MBConv layers, as well as on progressive learning methods. It equally uses a batch normalization and activation functions to enhance the extraction of features.

EfficientNetV2-S can perform well using less time to train but the resilience to distortions of images was not well studied in literature.

4.2.4 EfficientNet-B3

The main model employed in the given research is EfficientNet-B3 because the original FlexibleNet-B0 is worse in scaling and feature extraction. It relies on compound scaling to scale-up network depth, width and input resolution to enable representation of complex patterns.

In this paper, EfficientNet-B3 is started with ImageNet pre-trained weights, and the last classification layer is adjusted to five weather categories. The model is trained with the help of transfer learning and deeper layers are fine-tuned to enhance the results.

Also, EfficientNet-B3 is tested on various image variants including blur and color space conversion, and thus it is convenient to analyze its robustness in real-life situations.

4.3 Evaluation measures of proposed

The standard classification metrics that are based on the confusion matrix are used to assess the performance of the proposed model.

The performance of classification is estimated through the use of a confusion matrix that compares predicted and actual labels. It consists of:

TP: the number of correct positive predictions made by the model.

FP: the number of incorrect positive predictions made by the model.

TN: the number of correct negative predictions made by the model.

FN: the number of incorrect negative predictions made by the model.

Also there are also following statistical tools given the gender distribution of the underlying data: true positive rate (TPR), false positive rate (FPR), true negative rate (TNR), and false negative rate (FNR).

TPR: The ratio of correctly classified positive instances to the total actual positive instances.

$$TPR = \frac{TP}{TP + FN} * 100 \tag{2}$$

FPR: The ratio of incorrectly classified positive instances to the total actual negative instances.

$$FPR = \frac{FP}{FP + TN} * 100 \tag{3}$$

TNR: The ratio of correctly classified negative instances to the total actual negative instances.

$$TNR = \frac{TN}{TN + FP} * 100 \tag{4}$$

FNR: The ratio of incorrectly classified negative instances to the total actual positive instances.

$$FNR = \frac{FN}{FN + TP} * 100 \tag{5}$$

Precision: it measures a model's positive predictions. Hence, precision is the fraction of real positive predictions (i.e., accurately predicted positive samples) over the model's total positive predictions.

$$Precision = \frac{TP}{TP + FP} * 100 \tag{6}$$

Recall: it is commonly called as sensitivity or true positive rate, assesses a model's accuracy in identifying actual positives. True positives divided by the sum of true positives and false negatives is the recall.

$$Recall = \frac{TP}{TP + FN} * 100 \tag{7}$$

F1-Score: it combines precision and recall, is used to evaluate binary classification models.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} * 100 \tag{8}$$

Correct Classification Rate (CCR): It estimates the proportion of samples correctly classified. It is determined by dividing correct predictions by total predictions.

4.4 EfficientNet-B3 classification

The last classification model is EfficientNet-B3 because it to learn the intricate characterizations of features by scaling compounds. The model is trained in a multi-class dataset of five weather classes and is refined to obtain a better classification.

The model is also assessed based on various image variations during testing which includes blur image and color transformation. This enables one to study the behavior of the model in real-life conditions where the quality of an image can be different.

5. EXPERIMENTS AND RESULTS

5.1 Software and setup tools

Python programming language and deep learning models (e.g., TensorFlow/Keras and PyTorch) are used to achieve the proposed model. Google Colab, with GPU support, is used to run the experiments in order to speed up training. Image processing is done by using libraries like OpenCV, data processing and analysis is done by NumPy and Pandas libraries. Matplotlib and Seaborn are used in the visualization of results.

The hardware environment is GPU equipped resulting in massive time savings and enhanced computational efficiency. Training can also be made simpler using pre-trained models, which consume less resources.

5.2 Experimental dataset

The data sample applied in this research project is a public weather picture dataset accessed in Kaggle. It is a set of five weather types, which include cloudy, foggy, rainy, snowy and sunny. There are 18,038 in total of images used in the dataset which is divided into two groups to be trained and to be validated.

All the images are resized to a fixed input size before being trained into EfficientNet-B3. The data collection is systematized into class-based folders and it is loaded with image generation datasets. This makes sure that it is correctly labeled and the data is managed effectively in training.

5.3 Training and testing phase

In the dataset, an 80%-20% split is employed to split the dataset into training and validation sets. In the training step, ImageNet weights are loaded to pre-trained models after which it is fine-tuned to the task of weather classification. The last level of the classification is adjusted to produce five classes.

Multiple epochs of training are undertaken with either an optimizer like Adam, which is used with an appropriate learning rate. Training is enhanced by use of batch processing. Moreover, the weights on classes are also used in order to cope with class imbalance in the data.

At the validation stage performance is measured of the trained models on unseen data to determine their generalization abilities. Data are used to create predictions and calculate performance measures according to the confusion table.

5.4 Tuning classifier

During training, tuning of the classifier is done to have better model results. It comprises hyperparameter modification i.e., learning rate, batch size, and epochs. Unfreezing of the chosen layers of the pre-trained models is also done by fine-tuning enabling the network to be put into learning task-specific features.

Moreover, the adaptive learning rate and mixed precision training schemes are also applied to achieve efficiency in training. The layer of classifier is maximized to better class separation and misclassification between similar weather classes like the foggy and cloudy.

5.5 Result comparision and discussion

Table 1: Training and testing accuracy

Model	Training Accuracy (%)	Testing Accuracy (%)
EfficientNet-B0	98.86%	91.68%
DenseNet121	97.82%	91.88%
EfficientNetV2-S	98.53%	92.46%
EfficientNet-B3	95.89%	93.51%

The contention of training and testing accuracy of all four models is given in table 1. As can be seen, EfficientNet-B0 and EfficientNetV2-S are more accurate in the training than the other models. Nevertheless, they have a lower level of testing accuracy, which means that they tend to overfit.

DenseNet121 exhibits a little worse training accuracy yet similar testing accuracy. Conversely, EfficientNet-B3 has the best testing accuracy of 93.51 although its training accuracy is lower compared with other models. This is a sign of more generalization and less overfitting.

The findings indicate that although certain models are more aggressive in learning training data, EfficientNet-B3 offers a compromise between learning and generalization, which would be more able to serve in real-life weather classification problems.

Table 2: Classification using EfficientNet-B3 model

Model EfficientNet-B3	Precision (Avg)	Recall (Avg)	F1-Score (Avg)
Cloud	0.93	0.94	0.93
Foggy	0.91	0.93	0.92
Rainy	0.92	0.93	0.93
Snowy	0.97	0.93	0.95
Sunny	0.94	0.93	0.94

Table 2 indicates class performance of EfficientNet-B3 model relative to precision, recall and F1-score. The model exhibits a good and steady performance in all weather conditions.

The most successful one is the snowy class with 0.95 F1-score which means that the model is successful in identifying snowy backgrounds because of the clear visual characteristics of brightness and texture. High performance is demonstrated in the sunny and cloudy classes as well, although the precision and recall values are balanced.

The foggy class has a relatively low performance, with a F1-score of 0.92. This can be explained by the fact that the similar appearance of foggy and cloudy conditions complicates classification. In general, the findings show that EfficientNet-B3 can capture discriminative properties in most weather conditions and balance its performance.

Table 3: Applied color features

Variation	Cloudy	Sunny	Snowy	Rainy	Foggy
RGB	92.44	96.21	99.86	99.93	99.72
Blur_20	96.41	96.34	99.80	66.36	99.51
Blur_40	99.27	98.12	98.59	75.90	97.50
Blur_60	99.46	99.08	89.53	77.65	83.17
Grayscale	95.73	96.43	99.19	99.76	99.93
HSV	91.64	95.70	99.82	99.93	99.72
LAB	93.40	96.22	99.83	99.93	99.68
BGR	76.49	75.74	98.91	99.94	99.91
RGBA	92.44	96.21	99.86	99.93	99.72

Table 3 shows the robustness analysis of the proposed EfficientNet-B3 model in the view of multiple image transformations, such as blur, grayscale, and various color space representations. The performance is evaluated in terms of classification confidence (%) for each weather class.

Using the table, one can find that the model is consistently good on the original RGB images achieving high confidence on all the classes, snowy (99.86%), rainy (99.93%) and foggy (99.72) situations. This shows that the model has successfully learnt discriminative features using the dataset.

Under the case of Gaussian blur, the performance differs with the class. In cloudy and sunny images, the model has high confidence even at greater levels of blur, up to 99.46% and 99.08, respectively. Nevertheless, rainy pictures show a significant drop in performance with the confidence going down to 66.36% at Blur_20 and lower at higher blur values. This implies that rainy conditions are more or less dependent on fine-grained visual elements like streak patterns, which blur. Overall, similar to foggy and snowy classes, slightly weaker performance is observed at higher levels of blur, but nevertheless, the performance is rather high compared to rainy images.

The model is still very strong across all the classes in the case of grayscale transformation with a confidence value of above 95. This means that the model is not entirely based on information of color and can extract structural and texture based features effectively.

The model has both stable and high confidence levels especially when it comes to snowy, rainy and foggy classes having a confidence of over 99. This shows that the model is flexible to other color representations and generalizes well to other color distributions.

Nevertheless, there is a strong decrease in performance in the BGR colour space in cloudy (76.49) and sunny (75.74) classes. This shows that the improper color channel sequence can influence the capability of the model to appropriately decode visual features, particularly to those classes which require more color-based information like brightness and sky conditions.

The RGBA transformation produces the same results as RGB meaning that the extra alpha channel has no effect on the classification results when appropriately transformed.

In sum, the findings indicate the EfficientNet-B3 model is robust enough to numerous image changes, denoted in the grayscale and color space changes. It is, however, fragile to strong blur in some classes like rainy and inappropriate color channel setups like BGR. The results of the observations convey the significance of a proper representation of spatial information and color in case of optimal performance in weather image classification.

6. Conclusion

In this paper, a deep-learning based image-based weather classification system was created. It has utilized EfficientNet-B3 as the main model and compared it with EfficientNet-B0, DenseNet121, and EfficientNetV2-S to look at performance. The models were also trained and tested using a multi-class weather data which contained five categories; cloudy, fogdy, rainy, snowy and sunny.

The experimental findings show that EfficientNet-B3 is the most accurate to test as well as has better generalization than other models. Although other models were more training accurate, they exhibited overfitting behavior, and EfficientNet-B3 exhibited a more desirable balance between training and testing performance.

Besides accuracy, model robustness was also assessed in this study in the context of image variations like blur and color transformation. The findings point out that EfficientNet-B3 is not affected by different image conditions and, therefore, it can be more applicable in real-life scenarios where the quality of images can be changing.

On the whole, the offered strategy brings out the usefulness of EfficientNet-based structures in weather classification and the need to take into account both the presence and resilience when it comes to designing an effective architecture.

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